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4.4 Chemical Changes

AQA GCSE Chemistry (8462) — Paper 1

This topic is about three big families of reactions that turn up again and again across the whole course: the reactions of **metals** (and how we get metals out of their ores), the reactions of **acids** (and how we make salts and measure concentrations), and **electrolysis** (using electricity to split ionic compounds). A single idea links them all — **oxidation and reduction**. Once you can see a reaction in terms of which species loses electrons and which gains them, most of this topic falls into place.

Content marked **(HT)** is Higher Tier only. Content marked **(Triple)** is only examined in GCSE Chemistry (8462), not in Combined Science.

4.4.1 Reactivity of Metals

4.4.1.1 Metal Oxides — Oxidation and Reduction (Oxygen Definition)

Metals react with oxygen to produce **metal oxides**. Because oxygen is being added to the metal, these reactions are **oxidation** reactions.

magnesium + oxygen → magnesium oxide	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
iron + oxygen → iron(III) oxide	$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$
copper + oxygen → copper(II) oxide	$2Cu(s) + O_2(g) \rightarrow 2CuO(s)$

> **Oxidation** is the **gain of oxygen** by a substance.
> **Reduction** is the **loss of oxygen** from a substance.
>
> Oxidation and reduction always happen together — if one substance gains oxygen, another must have lost it. A reaction in which both happen is called a **redox** reaction.

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For example, when copper(II) oxide is heated with carbon, the copper oxide loses oxygen (it is **reduced**) and the carbon gains oxygen (it is **oxidised**):

```
2CuO(s) + C(s)  →  2Cu(s) + CO₂(g)

  CuO → Cu     loses oxygen  →  REDUCED
  C   → CO₂    gains oxygen  →  OXIDISED
```

The substance that does the reducing (here, carbon) is called the **reducing agent**; the substance that provides the oxygen (copper oxide) is the **oxidising agent**. You do not need these terms for AQA, but they make extended answers clearer.

> **Exam Tip:** At GCSE you meet *two* definitions of oxidation and reduction. At Foundation level you only need the oxygen definition (gain/loss of oxygen). At Higher tier you also need the electron definition (4.4.1.4). Read the question carefully — if it says "in terms of electrons", an answer about oxygen will score zero.

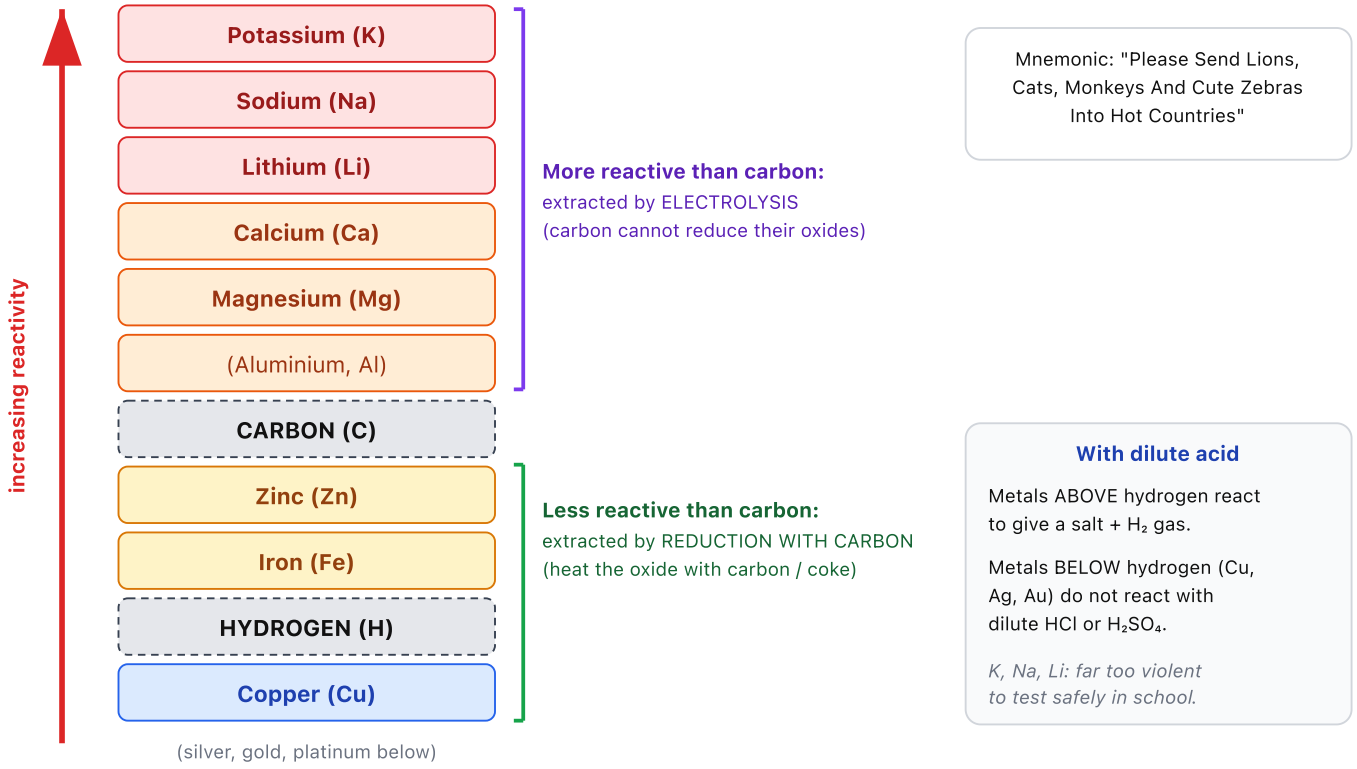
4.4.1.2 The Reactivity Series

When metals react with other substances, the metal atoms **lose electrons** to form **positive ions**. The **reactivity of a metal is related to its tendency to form positive ions**: the more easily a metal loses its outer electrons, the more reactive it is. This is why Group 1 metals (one outer electron, easily lost) are so much more reactive than transition metals such as copper.

Metals can be arranged in order of their reactivity, from their reactions with **water** and with **dilute acids**. The non-metals **carbon** and **hydrogen** are included in the series as reference points, because they are useful for predicting extraction methods and acid reactions.

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The reactivity series (AQA)



Reactions of Metals with Water

Reactive metals react with cold water to form a **metal hydroxide** and **hydrogen**:

```
metal + water → metal hydroxide + hydrogen

2Na(s) + 2H₂O(l) → 2NaOH(aq) + H₂(g)
Ca(s)  + 2H₂O(l) → Ca(OH)₂(aq) + H₂(g)
```

Less reactive metals such as magnesium react extremely slowly with cold water but react with **steam** to form the metal *oxide* and hydrogen: `Mg(s) + H₂O(g) → MgO(s) + H₂(g)`.

Metal	Reaction with cold water	Reaction with dilute hydrochloric acid
Potassium	Very vigorous. Floats, whizzes around, melts, hydrogen ignites with a **lilac** flame.	Explosive — not done in schools.

Sodium	Vigorous. Floats, melts into a silvery ball, fizzes and moves around. May ignite (orange flame).	Explosive — not done in schools.
Lithium	Floats and fizzes steadily, gradually disappears. Does not melt.	Very vigorous — not done in schools.
Calcium	Fizzes steadily, sinks; solution goes cloudy (calcium hydroxide is only slightly soluble).	Very vigorous fizzing.
Magnesium	Very slow — a few bubbles after several days. Reacts readily with steam.	Vigorous fizzing; metal disappears quickly; tube gets warm.
Zinc	No visible reaction with cold water.	Steady fizzing.
Iron	No visible reaction (rusts very slowly over days in the presence of oxygen).	Slow fizzing, a few bubbles.
Copper	No reaction.	No reaction.

Exam Tip: When describing observations, say what you would *see*: "fizzing / effervescence", "metal gets smaller and disappears", "floats", "melts into a ball", "lilac flame". "Hydrogen is produced" is not an observation — you cannot see a gas's identity. The test for hydrogen is a **lit splint** held at the mouth of the tube, which goes out with a **squeaky pop**.

Displacement Reactions

A **displacement reaction** is one in which a **more reactive** metal takes the place of a **less reactive** metal in a compound. The more reactive metal ends up as ions in the compound; the less reactive metal is released as the element.

```
magnesium + copper(II) sulfate → magnesium sulfate + copper
Mg(s)     + CuSO₄(aq)          → MgSO₄(aq)         + Cu(s)
```

```
Observations: blue solution fades to colourless; a pink-brown
coating of copper forms on the magnesium; the mixture warms up.

copper + magnesium sulfate → NO REACTION
(copper is less reactive than magnesium, so it cannot displace it)
```

Displacement experiments are used to **deduce an order of reactivity**. Each metal is placed in solutions of the salts of the other metals, and a reaction is recorded as a tick. The metal that displaces the most others is the most reactive. Displacement reactions are also exothermic — the bigger the gap in reactivity between the two metals, the bigger the temperature rise.

Metal added ↓ / Solution →	Magnesium sulfate	Zinc sulfate	Iron(II) sulfate	Copper(II) sulfate
Magnesium	—	✓ grey coating	✓ dark coating	✓ pink-brown coating
Zinc	×	—	✓	✓
Iron	×	×	—	✓
Copper	×	×	×	—

Number of reactions: Mg 3, Zn 2, Fe 1, Cu 0 → order of reactivity: **Mg > Zn > Fe > Cu**.

You should also be able to **deduce an order of reactivity from experimental results** such as the rate of bubbling with acid or the temperature rise in a displacement reaction, and to **explain** why a reaction does or does not happen using the series.

4.4.1.3 Extraction of Metals and Reduction

Unreactive metals such as gold (and some silver and platinum) are found in the Earth's crust as the **uncombined element** — they are said to occur "native". They can be separated from the surrounding rock by physical methods such as panning. Most metals, however, are found as **compounds** (usually oxides, sulfides or carbonates) in rocks called **ores**. A chemical reaction is needed to extract the metal.

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An **ore** is a rock that contains enough of a metal compound to make it economically worthwhile to extract the metal.

Metals less reactive than carbon can be extracted from their oxides by **reduction with carbon**: the carbon removes the oxygen from the metal oxide.
Metals more reactive than carbon must be extracted by **electrolysis** of the molten compound, which is much more expensive because of the large amount of electricity needed.

```
Extraction of iron in a blast furnace (simplified)

  iron(III) oxide + carbon → iron + carbon dioxide
  2Fe₂O₃(s)       + 3C(s)  → 4Fe(l) + 3CO₂(g)

  In practice the main reducing agent is carbon monoxide:
  Fe₂O₃(s) + 3CO(g) → 2Fe(l) + 3CO₂(g)

  Fe₂O₃ → Fe   loses oxygen → reduced
  C / CO → CO₂ gains oxygen → oxidised

Extraction of zinc
  2ZnO(s) + C(s) → 2Zn(s) + CO₂(g)
```

Metal	Position relative to carbon	Extraction method	Comment
K, Na, Li, Ca, Mg, Al	Above carbon	Electrolysis of molten compound	Expensive — large energy cost for melting and for the electric current.
Zn, Fe, Sn, Pb, Cu	Below carbon	Heat oxide with carbon (reduction)	Cheaper; carbon (as coke) is abundant. Copper is also extracted by other methods (see 4.10).

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Ag, Au, Pt	Very unreactive	Found native; separated physically	Gold may be found as nuggets or flakes in rock and river gravel.

Exam Tip: The question "Explain why aluminium cannot be extracted by heating its oxide with carbon" needs the comparison: "Aluminium is **more reactive than carbon**, so carbon cannot remove the oxygen from (reduce) aluminium oxide." Just saying "aluminium is very reactive" is not enough — you must compare it with carbon.

4.4.1.4 Oxidation and Reduction in Terms of Electrons (HT)

Oxidation is the loss of electrons.
Reduction is the gain of electrons.

Remember: **OIL RIG** — **O**xidation **I**s **L**oss, **R**eduction **I**s **G**ain (of electrons).

Higher-tier students must be able to write **ionic equations** for displacement reactions and identify in a symbol equation which species are oxidised and which are reduced. The trick is to remove **spectator ions** — ions that are present on both sides unchanged and take no part in the reaction.

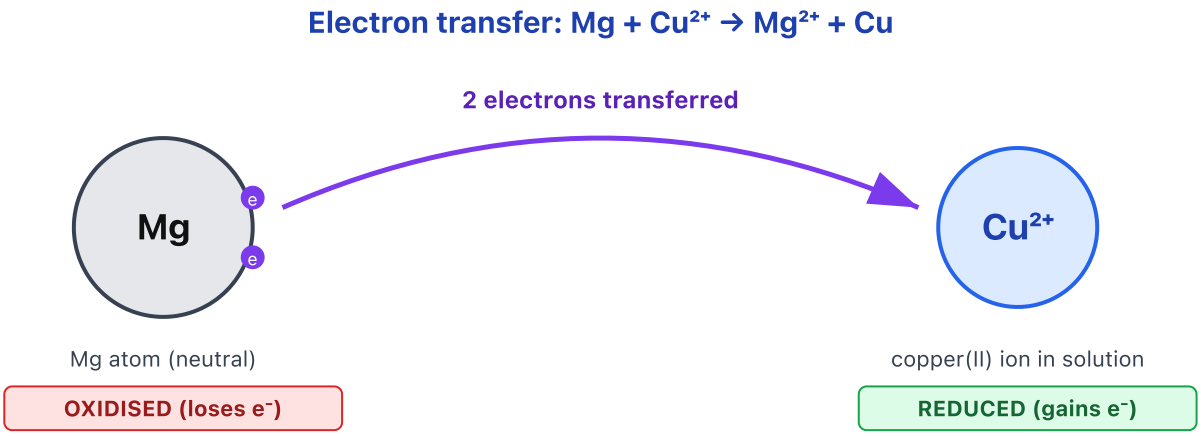
```
Full equation:   Mg(s) + CuSO₄(aq) → MgSO₄(aq) + Cu(s)

Split into ions: Mg(s) + Cu²⁺(aq) + SO₄²⁻(aq) → Mg²⁺(aq) + SO₄²⁻(aq) + Cu(s)

SO₄²⁻ appears unchanged on both sides → spectator ion → cancel it.

Ionic equation:  Mg(s) + Cu²⁺(aq) → Mg²⁺(aq) + Cu(s)

Half equations:
  Mg → Mg²⁺ + 2e⁻          Mg LOSES electrons  → OXIDISED
  Cu²⁺ + 2e⁻ → Cu          Cu²⁺ GAINS electrons → REDUCED
```



Worked Example 1 — Ionic Equation and Half Equations (HT)

```
Question: Zinc reacts with iron(II) sulfate solution.
          (a) Write the ionic equation.
          (b) Write half equations and identify the species
              oxidised and reduced.

(a) Zn(s) + FeSO₄(aq) → ZnSO₄(aq) + Fe(s)
    SO₄²⁻ is a spectator ion.
    Ionic: Zn(s) + Fe²⁺(aq) → Zn²⁺(aq) + Fe(s)

(b) Zn  → Zn²⁺ + 2e⁻     zinc atoms lose electrons: OXIDISED
    Fe²⁺ + 2e⁻ → Fe      iron(II) ions gain electrons: REDUCED

Check: charges balance on both sides of each half equation
       (0 = +2 − 2 and +2 − 2 = 0).
```

Exam Tip: In half equations, electrons are written as `e⁻`. Always check that the **total charge** on each side is the same — this is the quickest way to spot a missing electron. You may also see oxidation written as `Mg − 2e⁻ → Mg²⁺`; AQA accepts this form, but adding electrons to the right-hand side is clearer.

4.4.2 Reactions of Acids

4.4.2.1 Reactions of Acids with Metals

Acids react with metals that are **above hydrogen** in the reactivity series to produce a **salt** and **hydrogen**:

> **metal + acid → salt + hydrogen**

```
magnesium + hydrochloric acid → magnesium chloride + hydrogen
Mg(s)     + 2HCl(aq)          → MgCl₂(aq)          + H₂(g)

zinc + sulfuric acid → zinc sulfate + hydrogen
Zn(s) + H₂SO₄(aq)    → ZnSO₄(aq)    + H₂(g)

iron + hydrochloric acid → iron(II) chloride + hydrogen
Fe(s) + 2HCl(aq)         → FeCl₂(aq)         + H₂(g)
```

Knowledge of reactions limited to **magnesium, zinc and iron** with **hydrochloric and sulfuric acids** is expected. Copper does not react because it is below hydrogen.

Acid–Metal Reactions as Redox (HT)

Higher-tier students should be able to explain *in terms of gain or loss of electrons* that these are redox reactions, and identify which species are oxidised and which are reduced.

```
Mg(s) + 2HCl(aq) → MgCl₂(aq) + H₂(g)

Ions:   Mg + 2H⁺ + 2Cl⁻ → Mg²⁺ + 2Cl⁻ + H₂
Cl⁻ is a spectator ion.

Ionic:  Mg(s) + 2H⁺(aq) → Mg²⁺(aq) + H₂(g)

Mg   → Mg²⁺ + 2e⁻     magnesium is OXIDISED (loses electrons)
2H⁺ + 2e⁻ → H₂        hydrogen ions are REDUCED (gain electrons)
```

4.4.2.2 Neutralisation of Acids and Salt Production

Acids are neutralised by **alkalis** (soluble metal hydroxides) and **bases** (insoluble metal hydroxides and metal oxides) to produce a **salt and water**, and by **metal carbonates** to produce a **salt, water and carbon dioxide**.

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A **base** is a substance that neutralises an acid to form a salt and water (e.g. metal oxides and metal hydroxides).
An **alkali** is a base that **dissolves in water**, producing hydroxide ions, $OH^-(aq)$ (e.g. sodium hydroxide, potassium hydroxide).
A **salt** is the compound formed when the hydrogen in an acid is replaced by a metal (or by the ammonium ion).

Reaction type	General equation	Example
Acid + alkali	acid + metal hydroxide → salt + water	$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
Acid + base (oxide)	acid + metal oxide → salt + water	$H_2SO_4(aq) + CuO(s) \rightarrow CuSO_4(aq) + H_2O(l)$
Acid + base (insoluble hydroxide)	acid + metal hydroxide → salt + water	$2HNO_3(aq) + Cu(OH)_2(s) \rightarrow Cu(NO_3)_2(aq) + 2H_2O(l)$
Acid + carbonate	acid + metal carbonate → salt + water + carbon dioxide	$2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$
Acid + metal	acid + metal → salt + hydrogen	$2HCl(aq) + Zn(s) \rightarrow ZnCl_2(aq) + H_2(g)$

Naming Salts

The **first part** of a salt's name comes from the **metal** (or ammonium) in the base, alkali or carbonate. The **second part** comes from the **acid**:

Acid	Formula	Salt produced	Anion	Example salt
Hydrochloric acid	HCl	… **chloride**	Cl^-	sodium chloride, $NaCl$
Sulfuric acid	H_2SO_4	… **sulfate**	SO_4^{2-}	copper(II) sulfate, $CuSO_4$
Nitric acid	HNO_3	… **nitrate**	NO_3^-	potassium nitrate, KNO_3

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You must be able to **predict products** from given reactants and **use the formulae of common ions** to deduce the formulae of salts. Balance the charges: for example, aluminium (Al^{3+}) with sulfate (SO_4^{2-}) gives $Al_2(SO_4)_3$ ($2 \times +3 = +6$; $3 \times -2 = -6$).

> **Exam Tip:** The test for carbon dioxide is to bubble the gas through **limewater** (calcium hydroxide solution), which turns **milky / cloudy**. If a question describes "fizzing" when an acid reacts with a solid that is *not* a metal, the solid is almost certainly a carbonate and the gas is CO_2.

4.4.2.3 Soluble Salts

Soluble salts can be made from acids by reacting them with **solid insoluble substances** such as metals, metal oxides, hydroxides or carbonates. The solid is added to the acid until **no more reacts** (the solid is in **excess**). The excess solid is **filtered off** to leave a solution of the salt only. The salt solution is then **crystallised** to produce solid salt.

Required Practical 1 — Preparing a Pure, Dry Sample of a Soluble Salt

Aim: to prepare a pure, dry sample of a soluble salt (e.g. copper(II) sulfate) from an insoluble oxide or carbonate (e.g. copper(II) oxide), using a Bunsen burner to heat dilute acid and a water bath or electric heater to evaporate the solution.

```
CuO(s) + H₂SO₄(aq) → CuSO₄(aq) + H₂O(l)
black     colourless     blue

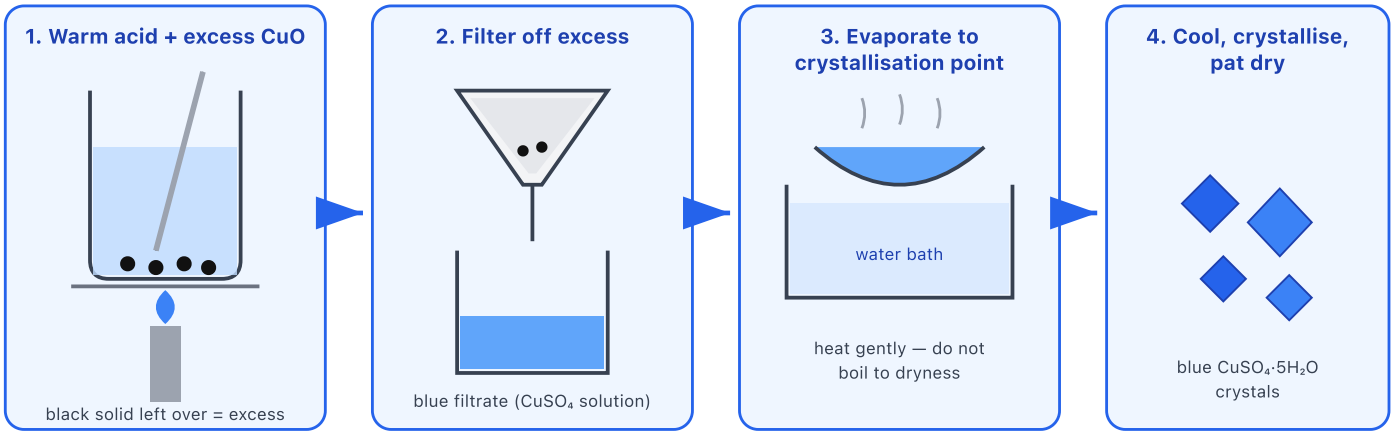
Crystals: CuSO₄·5H₂O (blue, hydrated copper(II) sulfate)
```

1. Measure about **40 cm³ of dilute sulfuric acid** into a beaker using a measuring cylinder.
2. **Warm the acid gently** with a Bunsen burner (do not boil) — this speeds up the reaction.
3. Add **copper(II) oxide** a spatula at a time, stirring with a glass rod, until **no more dissolves** and some black solid remains. This shows the copper oxide is in **excess**, so all the acid has been used up.
4. **Filter** the warm mixture through filter paper in a funnel into an evaporating basin, to remove the unreacted copper oxide. The filtrate is a blue solution of copper(II) sulfate.

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5. **Heat the solution** gently over a water bath (or with an electric heater) to evaporate some of the water until the **crystallisation point** is reached — when a drop placed on a cold glass rod forms crystals, or crystals start to form at the edge of the solution. Do not boil to dryness.
6. Leave the solution in a **cool place** to crystallise slowly (giving larger crystals).
7. Pour off any remaining liquid, then **pat the crystals dry** between sheets of filter paper (or leave them in a warm place / desiccator).

Making copper(II) sulfate crystals



Step	Reason
Warm the acid	Increases the rate of reaction so the base dissolves faster.
Add base in excess	Ensures **all the acid reacts**, so the final salt is not contaminated with acid.
Filter	Removes the unreacted (insoluble) excess base.
Evaporate only to the crystallisation point	Boiling to dryness can decompose the salt, drive off water of crystallisation (giving white anhydrous powder), and causes spitting.
Cool slowly	Produces larger, purer crystals.
Pat dry with filter paper	Removes surface water without heating (which could dehydrate the crystals).

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Hazards: sulfuric acid is an irritant/corrosive — wear eye protection. Copper(II) oxide and copper(II) sulfate are harmful and dangerous to the environment — do not pour down the sink. Hot apparatus can burn — allow to cool before handling.

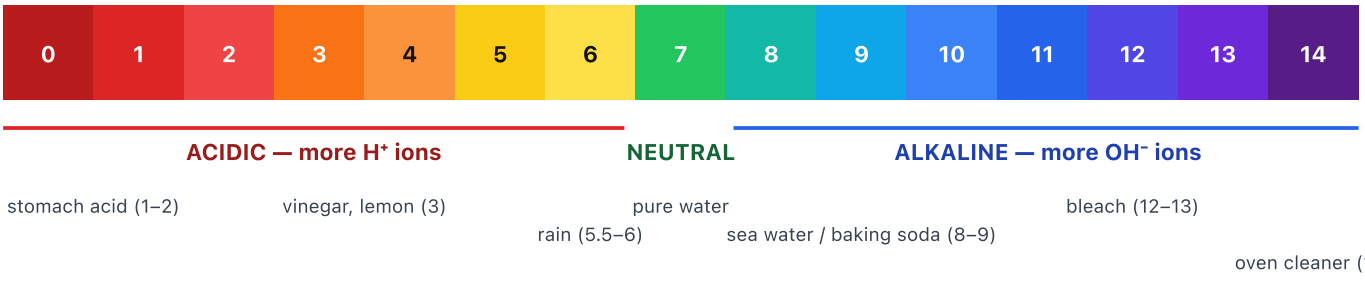
> **Exam Tip:** Six-mark method questions on this practical are marked on a logical sequence: **warm acid → add solid until in excess → filter → heat filtrate to crystallisation point → leave to cool → dry crystals**. Name the specific apparatus (filter funnel and paper, evaporating basin, water bath) and give the reason for adding the solid in excess. Students often lose marks by evaporating "until all the water has gone".

4.4.2.4 The pH Scale and Neutralisation

> **Acids** produce **hydrogen ions, $H^+(aq)$**, in aqueous solutions.
> **Aqueous solutions of alkalis** contain **hydroxide ions, $OH^-(aq)$**.
>
> The **pH scale**, from 0 to 14, is a measure of the acidity or alkalinity of a solution. A solution with **pH 7 is neutral**; pH less than 7 is **acidic**; pH greater than 7 is **alkaline**.

pH can be measured using **universal indicator** (which gives a colour that is compared with a chart) or a **pH probe / pH meter** (which gives a numerical reading, often to 1 or 2 decimal places, and is more accurate and less subjective).

The pH scale and universal indicator colours

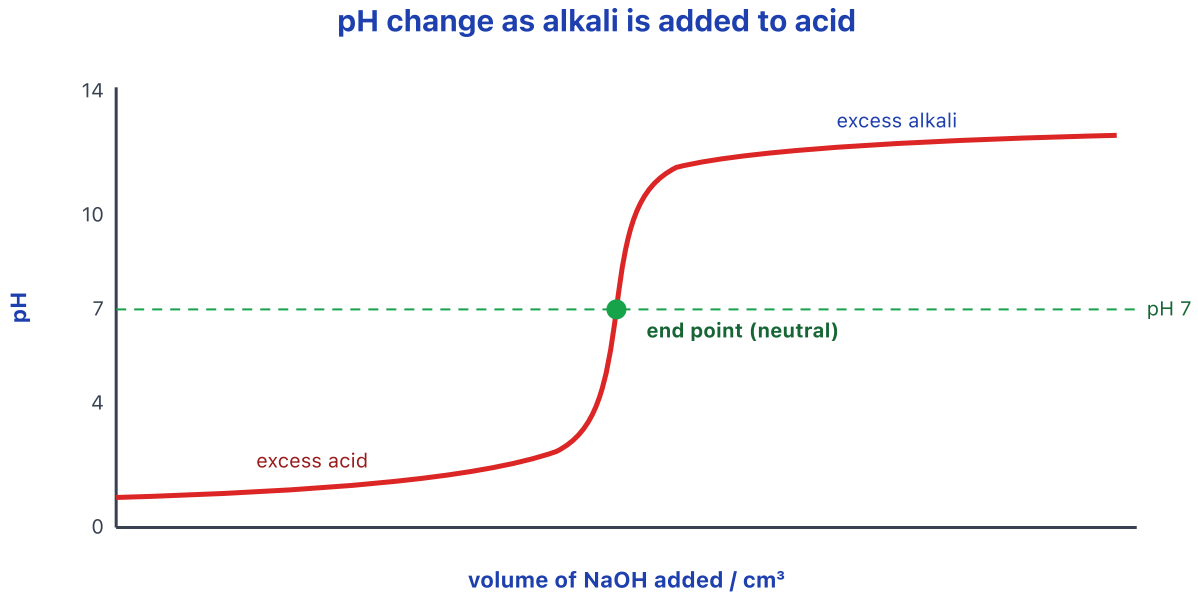


In **neutralisation** reactions between an acid and an alkali, hydrogen ions react with hydroxide ions to produce water. This is the **ionic equation for every acid–alkali neutralisation**:

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

The metal ion from the alkali and the non-metal ion from the acid are spectator ions; they remain in solution as the salt.

You should be able to describe the use of universal indicator or a wide-range indicator to measure the approximate pH of a solution, and to use the pH scale to identify acidic or alkaline solutions. In a neutralisation curve, adding alkali to acid causes the pH to rise slowly at first, then very sharply around pH 7 (the **end point**), then level off.



4.4.2.5 Titrations (Triple)

The volumes of acid and alkali solutions that react with each other can be measured by **titration** using a suitable indicator. If the concentration of one solution is known, the concentration of the other can be calculated.

A **titration** is a technique used to find the exact volume of one solution that reacts completely with a known volume of another solution. The point at which the reaction is just complete is called the **end point**, shown by a sharp colour change of the indicator.

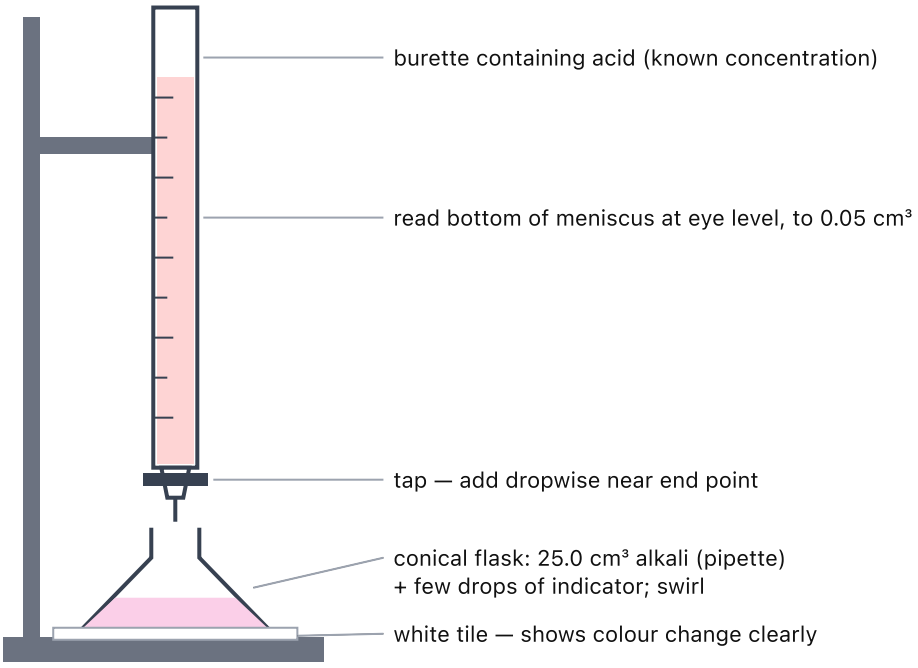
Required Practical 2 — Titration (Triple)

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Aim: to determine the reacting volumes of solutions of a strong acid and a strong alkali by titration, and (HT) to determine the concentration of one of the solutions in mol/dm³ and g/dm³ from the reacting volumes and the known concentration of the other solution.

1. Rinse a **25.0 cm³ volumetric pipette** with the alkali, then use it (with a pipette filler) to transfer exactly 25.0 cm³ of sodium hydroxide solution into a clean **conical flask**.
2. Add a few drops of a suitable **indicator** — e.g. **phenolphthalein** (pink in alkali, colourless in acid) or **methyl orange** (yellow in alkali, red in acid).
3. Rinse a **burette** with the acid, then fill it with acid using a small funnel. Remove the funnel. Run a little acid through the tap to fill the jet. Record the initial reading to the nearest **0.05 cm³**, reading from the **bottom of the meniscus** at eye level.
4. Place the flask on a **white tile** (so the colour change is easy to see). Add acid from the burette, swirling the flask constantly.
5. As the end point approaches (the colour lingers longer where the acid enters), add the acid **dropwise** until the indicator changes colour permanently. Record the final burette reading.
6. The first run is a **rough titration**. Repeat accurately until you have **concordant results** — titres within **0.10 cm³** of each other.
7. Calculate the **mean titre** using only the concordant results.

Titration apparatus



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	Rough	Run 1	Run 2	Run 3
Final reading / cm³	23.20	45.65	22.50	45.00
Initial reading / cm³	0.00	23.20	0.00	22.50
Titre / cm³	23.20	**22.45**	**22.50**	**22.50**

Concordant titres (within 0.10 cm³): 22.45, 22.50, 22.50 → mean titre = (22.45 + 22.50 + 22.50) ÷ 3 = **22.48 cm³**.

> **Exam Tip:** Do **not** use universal indicator in a titration — it changes colour gradually through many colours, so there is no sharp end point. Use a single indicator such as phenolphthalein or methyl orange. Also, the rough titration is *never* included in the mean.

Titration Calculations (HT, Triple)

> `moles (mol) = concentration (mol/dm³) × volume (dm³)`
>
> `concentration (g/dm³) = concentration (mol/dm³) × $M_r$`
>
> To convert cm³ to dm³, **divide by 1000** (1 dm³ = 1000 cm³).

The four-step method for any titration calculation:

1. Calculate the **moles** of the substance whose concentration and volume you **know**.
2. Use the **balanced equation** (mole ratio) to find the moles of the other substance.
3. Calculate the **concentration** of the unknown in mol/dm³ = moles ÷ volume in dm³.
4. If asked, convert to **g/dm³** by multiplying by M_r.

Worked Example 2 — 1 : 1 Titration (HT)

```
Question: 25.0 cm³ of sodium hydroxide solution was neutralised by
          22.50 cm³ of 0.100 mol/dm³ hydrochloric acid.
```

```
          Calculate the concentration of the NaOH in (a) mol/dm³
          and (b) g/dm³.  (Mr NaOH = 40)

Equation: HCl + NaOH → NaCl + H₂O       ratio 1 : 1

Step 1:  moles HCl = 0.100 × (22.50 ÷ 1000)
                   = 0.100 × 0.02250
                   = 0.002250 mol
Step 2:  moles NaOH = 0.002250 mol  (1 : 1 ratio)
Step 3:  conc NaOH = 0.002250 ÷ (25.0 ÷ 1000)
                   = 0.002250 ÷ 0.0250
                   = 0.0900 mol/dm³
Step 4:  conc NaOH = 0.0900 × 40
                   = 3.60 g/dm³

Answer:  (a) 0.0900 mol/dm³   (b) 3.60 g/dm³
```

Worked Example 3 — 2 : 1 Titration with Sulfuric Acid (HT)

```
Question: 25.0 cm³ of 0.200 mol/dm³ sodium hydroxide was neutralised
          by 20.00 cm³ of sulfuric acid. Find the concentration of the
          sulfuric acid in mol/dm³ and g/dm³.  (Mr H₂SO₄ = 98)

Equation: H₂SO₄ + 2NaOH → Na₂SO₄ + 2H₂O    ratio 1 : 2

Step 1:  moles NaOH = 0.200 × 0.0250 = 0.00500 mol
Step 2:  moles H₂SO₄ = 0.00500 ÷ 2   = 0.00250 mol
Step 3:  conc H₂SO₄  = 0.00250 ÷ 0.02000
                     = 0.125 mol/dm³
Step 4:  in g/dm³    = 0.125 × 98 = 12.25 g/dm³ ≈ 12.3 g/dm³

Answer:  0.125 mol/dm³;  12.3 g/dm³ (3 s.f.)
```

Exam Tip: The most common error in titration calculations is forgetting the **mole ratio**. Sulfuric acid is *diprotic* — one H_2SO_4 releases two H^+ ions and reacts with **two** NaOH. Always write the balanced equation first. The second most common error is forgetting to convert cm³ to dm³.

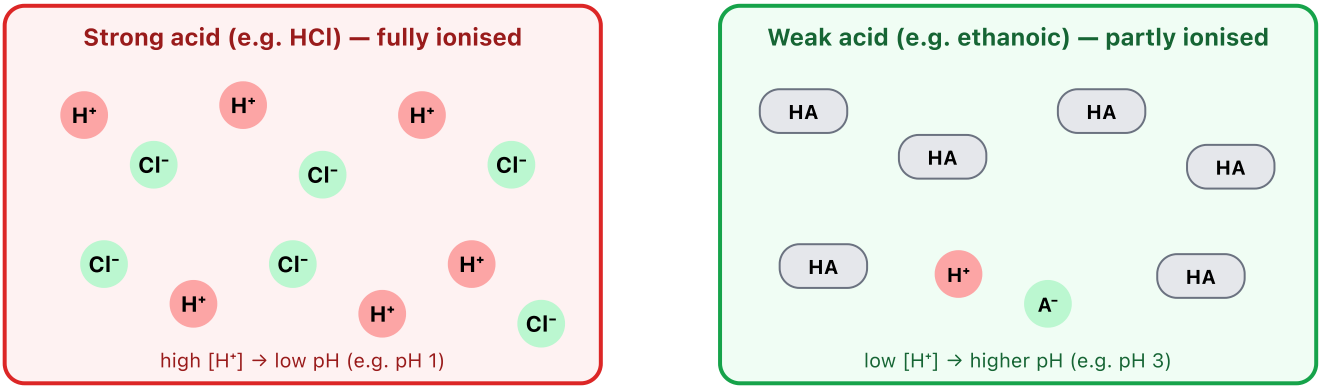
4.4.2.6 Strong and Weak Acids (HT)

A **strong acid** is **completely ionised** in aqueous solution. Examples: hydrochloric, nitric and sulfuric acids.

A **weak acid** is only **partially ionised** in aqueous solution. Examples: ethanoic, citric and carbonic acids.

```
Strong:  HCl(aq)          → H⁺(aq) + Cl⁻(aq)          (every molecule ionises)
Weak:    CH₃COOH(aq)      ⇌ H⁺(aq) + CH₃COO⁻(aq)      (only a small fraction i
                                                        reversible reaction)
```

Strong vs weak acid (same concentration)



Strength vs Concentration

"Strong/weak" and "concentrated/dilute" describe **two completely different things**:

Term	Meaning	Example
Strong / weak	The **proportion** of acid molecules that **ionise** in water.	HCl is strong at any concentration; ethanoic acid is weak at any concentration.
Concentrated / dilute	The **amount of acid** (in moles or grams) dissolved in a given volume of solution.	2.0 mol/dm³ is more concentrated than 0.1 mol/dm³.

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So it is perfectly possible to have a **dilute solution of a strong acid** (e.g. 0.01 mol/dm³ HCl) or a **concentrated solution of a weak acid** (e.g. 5 mol/dm³ ethanoic acid).

pH and Hydrogen Ion Concentration

> For a given concentration of aqueous solutions, the **stronger the acid**, the **lower the pH**.
>
> As the pH **decreases by one unit**, the hydrogen ion concentration of the solution **increases by a factor of 10**.

pH	$[H^+]$ / mol/dm³	Relative to pH 4
1	0.1 (10^{-1})	1000 × more H^+
2	0.01 (10^{-2})	100 × more H^+
3	0.001 (10^{-3})	10 × more H^+
4	0.0001 (10^{-4})	—

Worked Example 4 — pH and $[H^+]$ (HT)

```
Question: Solution A has pH 2. Solution B has pH 5.
          How many times greater is the hydrogen ion
          concentration in A than in B?

Difference in pH = 5 − 2 = 3 units
Factor           = 10³ = 1000

Answer:  A has 1000 times the H⁺ concentration of B.

Follow-up: 10 cm³ of an acid at pH 3 is diluted to 100 cm³
           with water. What is the new pH?
  [H⁺] falls by a factor of 10 → pH rises by 1 → pH 4
```

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Exam Tip: If a question asks you to explain why two acids of the **same concentration** have different pH values, the answer must mention **ionisation**: "Hydrochloric acid is a strong acid and is fully ionised, whereas ethanoic acid is a weak acid and is only partially ionised. So the concentration of H^+ ions is higher in hydrochloric acid, giving a lower pH." The same idea explains why a weak acid reacts more slowly with magnesium than a strong acid of the same concentration.

4.4.3 Electrolysis

4.4.3.1 The Process of Electrolysis

When an ionic compound is **melted** or **dissolved in water**, the ions are **free to move** about within the liquid or solution. These liquids and solutions are able to conduct electricity and are called **electrolytes**. In the solid state, the ions are held in fixed positions in the lattice and cannot move, so solid ionic compounds do not conduct.

Electrolysis is the process of using electricity to **decompose** an ionic compound (the electrolyte) into its elements, by passing a direct current through it when molten or in solution.

Cathode = the **negative** electrode. Positive ions (**cations**) move towards it.
Anode = the **positive** electrode. Negative ions (**anions**) move towards it.

At the electrodes, the ions are **discharged** — they gain or lose electrons to become neutral atoms or molecules:

- At the **cathode**, positive ions **gain** electrons — this is **reduction**.
- At the **anode**, negative ions **lose** electrons — this is **oxidation**.

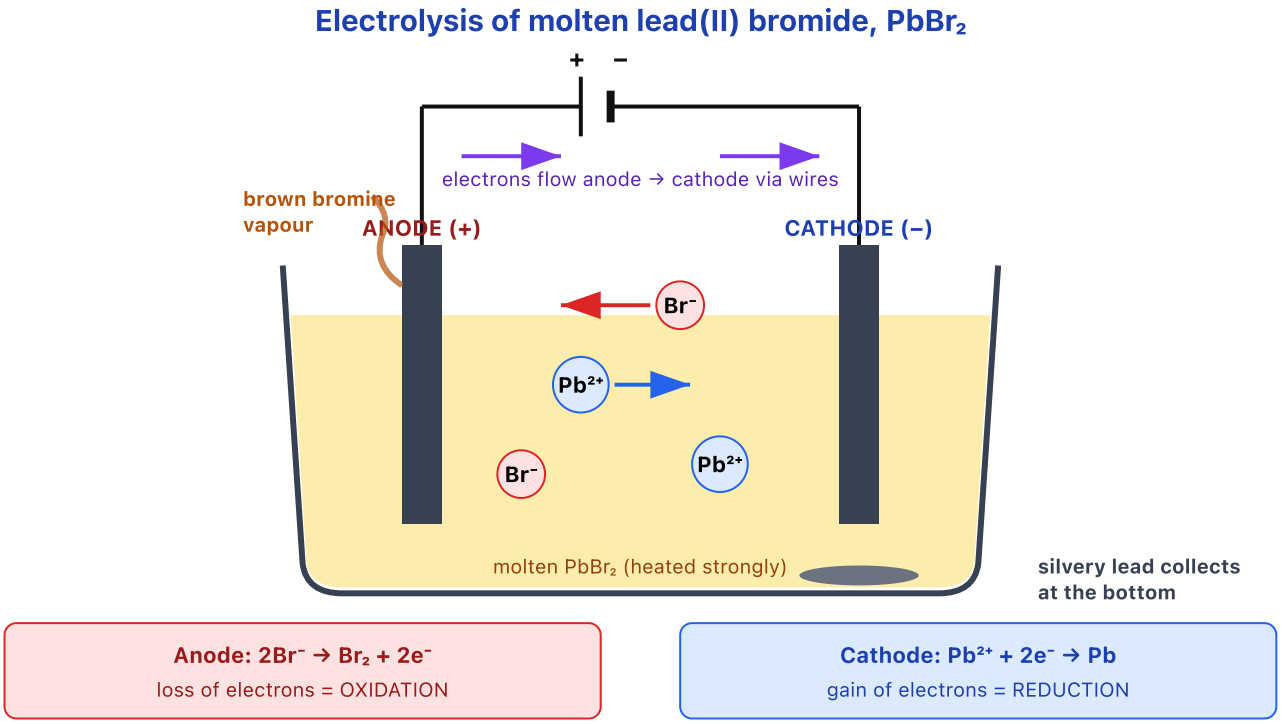
Electrodes are usually made of an **inert** material such as **graphite** or **platinum**, so they conduct electricity but do not take part in the reaction.

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Exam Tip: Use the memory aid **PANIC**: **P**ositive **A**node, **N**egative **I**s **C**athode. And **OIL RIG** links to it: oxidation at the anode (An Ox), reduction at the cathode (Red Cat). The electrons flow round the external circuit from the anode to the cathode — never through the electrolyte, where the current is carried by moving ions.

4.4.3.2 Electrolysis of Molten Ionic Compounds

When a **simple ionic compound** (e.g. lead bromide) is electrolysed in the molten state using inert electrodes, the **metal** (lead) is produced at the **cathode** and the **non-metal** (bromine) is produced at the **anode**. You should be able to predict the products of electrolysis of any binary ionic compound in the molten state.



Molten electrolyte	Product at cathode (−)	Product at anode (+)
Lead(II) bromide, $PbBr_2$	Lead	Bromine
Sodium chloride, NaCl	Sodium	Chlorine
Zinc chloride, $ZnCl_2$	Zinc	Chlorine

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Potassium iodide, KI	Potassium	Iodine
Aluminium oxide, Al_2O_3	Aluminium	Oxygen
Calcium bromide, $CaBr_2$	Calcium	Bromine

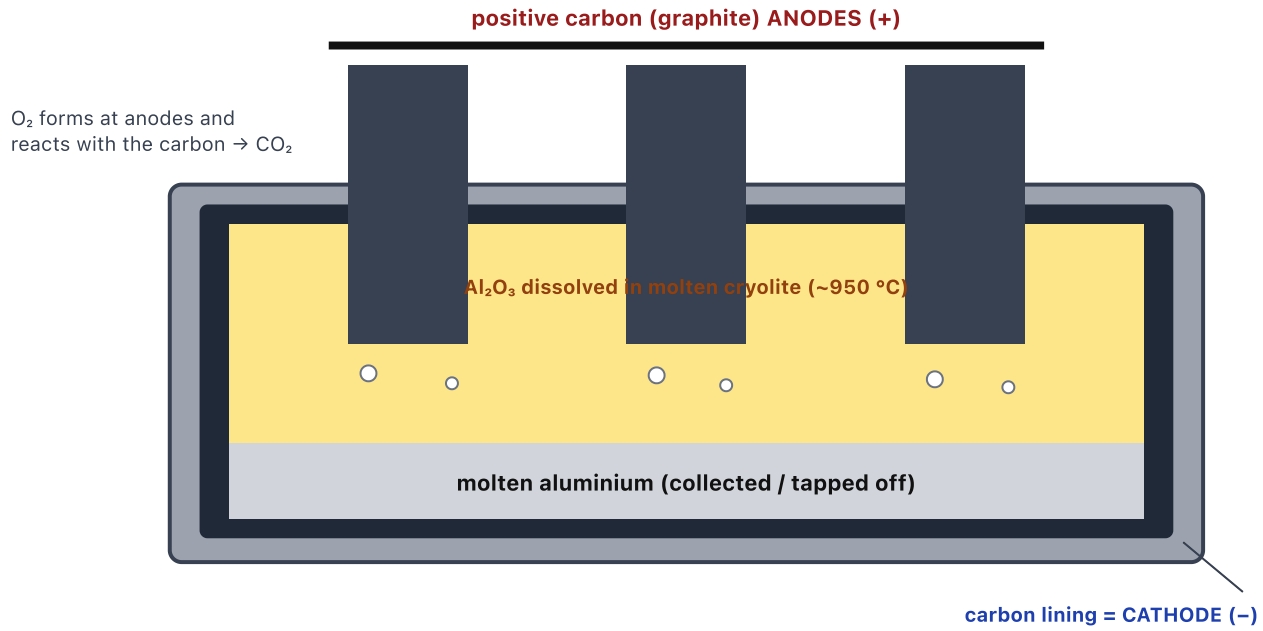
This experiment should be done in a fume cupboard because bromine vapour is toxic. Heat is needed to keep the electrolyte molten throughout.

4.4.3.3 Using Electrolysis to Extract Metals

Metals can be extracted from molten compounds using electrolysis. Electrolysis is used if the metal is **too reactive to be extracted by reduction with carbon**, or if the metal **reacts with carbon**. Large amounts of energy are used in the extraction process to **melt the compounds** and to **produce the electrical current**, which makes it expensive.

Aluminium is manufactured by the electrolysis of a molten mixture of **aluminium oxide** (from the ore bauxite) and **cryolite**, using **carbon** as the positive electrode (anode).

Extraction of aluminium by electrolysis



Feature	Explanation

Why electrolysis?	Aluminium is more reactive than carbon, so its oxide cannot be reduced by carbon.
Why cryolite?	Pure aluminium oxide has a very high melting point (over 2000 °C). Dissolving it in molten cryolite **lowers the operating temperature** to about 950 °C, **saving energy** and reducing costs. The molten mixture also conducts better.
Cathode reaction	$Al^{3+} + 3e^- \rightarrow Al$ (reduction) — molten aluminium sinks and is tapped off.
Anode reaction	$2O^{2-} \rightarrow O_2 + 4e^-$ (oxidation).
Why must the anode be replaced?	At the high temperature, the oxygen produced **reacts with the carbon anode** to form carbon dioxide ($C + O_2 \rightarrow CO_2$). The anodes gradually **burn away** and must be replaced regularly.
Overall reaction	$2Al_2O_3(l) \rightarrow 4Al(l) + 3O_2(g)$

Exam Tip: "Explain why the positive electrodes must be replaced" is a classic 3-mark question: (1) oxygen is produced at the positive electrode; (2) the oxygen reacts with the carbon; (3) forming carbon dioxide, so the electrode is used up. Also be ready to explain why aluminium is expensive: lots of electricity is needed both to keep the electrolyte molten and to provide the current.

4.4.3.4 Electrolysis of Aqueous Solutions

An aqueous solution contains the ions from the dissolved compound **and** a small number of **H^+** and **OH^-** ions from the partial ionisation of water: $H_2O \rightleftharpoons H^+ + OH^-$. The ions discharged at each electrode depend on the **relative reactivity** of the elements involved.

At the negative electrode (cathode): hydrogen is produced if the metal is **more reactive than hydrogen**. Otherwise, the metal is produced (e.g. copper, silver).

At the positive electrode (anode): oxygen is produced **unless** the solution contains **halide**

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ions (Cl^-, Br^-, I^-), in which case the **halogen** is produced. With sulfate, nitrate or carbonate solutions, oxygen is formed from hydroxide ions.

Predicting products of aqueous electrolysis



Ions NOT discharged stay in solution — e.g. electrolysis of brine leaves Na^+ and OH^-, so the remaining solution becomes sodium hydroxide.

Aqueous solution	Ions present	Cathode (–)	Anode (+)	Left in solution
Sodium chloride (brine)	Na^+, Cl^-, H^+, OH^-	Hydrogen	Chlorine	Sodium hydroxide
Copper(II) sulfate	Cu^{2+}, SO_4^{2-}, H^+, OH^-	Copper (pink deposit)	Oxygen	Sulfuric acid (blue colour fades)
Copper(II) chloride	Cu^{2+}, Cl^-, H^+, OH^-	Copper	Chlorine	Water
Potassium bromide	K^+, Br^-, H^+, OH^-	Hydrogen	Bromine (orange solution)	Potassium hydroxide
Sodium sulfate	Na^+, SO_4^{2-}, H^+, OH^-	Hydrogen	Oxygen	Sodium sulfate (water is electrolysed)
Dilute sulfuric acid	H^+, SO_4^{2-}, OH^-	Hydrogen	Oxygen	Sulfuric acid (more concentrated)

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Silver nitrate	Ag^+, NO_3^-, H^+, OH^-	Silver	Oxygen	Nitric acid

The electrolysis of **brine** (concentrated sodium chloride solution) is a major industrial process. All three products are useful: **chlorine** (for bleach, PVC and water treatment), **hydrogen** (for making ammonia and margarine, and as a fuel) and **sodium hydroxide** (for soap, paper and bleach manufacture).

Required Practical 3 — Electrolysis of Aqueous Solutions

Aim: to investigate what happens when aqueous solutions are electrolysed using **inert electrodes**. This should be an investigation involving developing a hypothesis.

1. Pour about 50 cm^3 of the solution (e.g. copper(II) sulfate, sodium chloride) into a beaker.
2. Place a lid (e.g. a Petri dish with two holes) on the beaker and push two **graphite rod electrodes** through the holes so that they do not touch each other.
3. Attach the electrodes to a **low-voltage DC power supply** (4–6 V) using crocodile clips and leads.
4. Turn on the supply and **observe** each electrode: is a gas produced, or a solid deposited?
5. Identify any gas using the appropriate test. Gases can be collected in small test tubes filled with the solution and inverted over the electrodes.
6. Repeat with other solutions and compare the products with your predictions.

Gas	Test	Positive result
Hydrogen	Hold a **lit (burning) splint** at the mouth of the tube	**Squeaky pop**
Oxygen	Insert a **glowing splint** into the tube	Splint **relights**
Chlorine	Hold **damp blue litmus paper** in the gas	Turns red, then is **bleached white**

Safety: chlorine is toxic — switch off the supply as soon as a positive result has been obtained, work in a well-ventilated room, and avoid inhaling. Copper(II) sulfate is harmful — wear eye

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protection. Keep voltages low.

> **Exam Tip:** In theory, equal volumes of hydrogen and chlorine are produced from brine. In practice, the volume of chlorine collected is **less** than hydrogen because chlorine is **slightly soluble in water** (and reacts with it). In the electrolysis of dilute sulfuric acid, the volume of hydrogen is **twice** the volume of oxygen — 4 electrons release 2 H_2 but only 1 O_2.

4.4.3.5 Representation of Reactions at Electrodes as Half Equations (HT)

During electrolysis, at the cathode (negative electrode), positively charged ions **gain electrons** and so the reactions are **reductions**. At the anode (positive electrode), negatively charged ions **lose electrons** and so the reactions are **oxidations**. These reactions can be represented by half equations.

Electrode	Ion discharged	Half equation	Type
Cathode	Hydrogen ion	$2H^+ + 2e^- \rightarrow H_2$	Reduction
Cathode	Copper(II) ion	$Cu^{2+} + 2e^- \rightarrow Cu$	Reduction
Cathode	Lead(II) ion	$Pb^{2+} + 2e^- \rightarrow Pb$	Reduction
Cathode	Sodium ion (molten)	$Na^+ + e^- \rightarrow Na$	Reduction
Cathode	Aluminium ion (molten)	$Al^{3+} + 3e^- \rightarrow Al$	Reduction
Cathode	Silver ion	$Ag^+ + e^- \rightarrow Ag$	Reduction
Anode	Chloride ion	$2Cl^- \rightarrow Cl_2 + 2e^-$	Oxidation
Anode	Bromide ion	$2Br^- \rightarrow Br_2 + 2e^-$	Oxidation
Anode	Iodide ion	$2I^- \rightarrow I_2 + 2e^-$	Oxidation
Anode	Oxide ion (molten)	$2O^{2-} \rightarrow O_2 + 4e^-$	Oxidation
Anode	Hydroxide ion	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	Oxidation

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Worked Example 5 — Writing Half Equations (HT)

Question: Aqueous copper(II) chloride is electrolysed with inert electrodes. Write half equations for the reactions at each electrode and state which is oxidation.

Ions present: Cu^{2+}, Cl^- (from $CuCl_2$); H^+, OH^- (from water)

Cathode (−): copper is less reactive than hydrogen → copper forms
$Cu^{2+} + 2e^- \rightarrow Cu$ (gain of e^- = REDUCTION)

Anode (+): halide present → chlorine forms
$2Cl^- \rightarrow Cl_2 + 2e^-$ (loss of e^- = OXIDATION)

Check balance: each half equation has equal atoms and equal charge on both sides (e.g. 2 × −1 = −2 = 0 − 2).

Worked Example 6 — Balancing the Hydroxide Half Equation (HT)

Start: $OH^- \rightarrow O_2$ need 2 O on the left... but H must go somewhere

Step 1: $4OH^- \rightarrow O_2 + 2H_2O$ atoms: 4 O and 4 H each side ✓
Step 2: charge left = −4, charge right = 0
add 4 electrons to the right to balance charge
Result: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

Alternative form accepted by AQA: $4OH^- - 4e^- \rightarrow O_2 + 2H_2O$

> **Exam Tip:** When asked "why is the reaction at the cathode a reduction?" the answer is "because the positive ions **gain electrons**". Do not say "because the ion gains oxygen" or "because it is at the negative electrode". The electron definition is the one rewarded in electrolysis questions.

Summary Checklist for 4.4 Chemical Changes

- Define oxidation and reduction in terms of gain/loss of **oxygen**, and (HT) gain/loss of **electrons** (OIL RIG).
- Recall the reactivity series (K, Na, Li, Ca, Mg, C, Zn, Fe, H, Cu) and link reactivity to the tendency to form positive ions.
- Describe the reactions of metals with water and dilute acids, and deduce an order of reactivity from observations.
- Predict and explain displacement reactions; (HT) write ionic and half equations, identify spectator ions and species oxidised/reduced.
- Explain how the position of a metal relative to carbon determines its extraction method; know that gold is found native.
- Write equations for acid + metal, acid + alkali/base, acid + carbonate; name the salts formed by HCl, H_2SO_4 and HNO_3.
- Describe Required Practical 1 — making a pure, dry soluble salt from an insoluble oxide or carbonate — with reasons for each step.
- Use the pH scale; recall that acids produce H^+ and alkalis produce OH^-; write $H^+ + OH^- \rightarrow H_2O$.
- (Triple) Describe Required Practical 2 — titration — including concordant results; (HT) calculate concentrations in mol/dm³ and g/dm³.
- (HT) Distinguish strong/weak from concentrated/dilute; explain that a pH decrease of 1 means a tenfold increase in $[H^+]$.
- Explain why ionic compounds conduct only when molten or dissolved; predict products of electrolysis of molten binary compounds.
- Describe the extraction of aluminium: cryolite, carbon anodes, why anodes are replaced, why it is expensive.
- Predict products of aqueous electrolysis (H_2 vs metal; O_2 vs halogen) and describe Required Practical 3 with gas tests.
- (HT) Write half equations at each electrode, including $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$.

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